

AEGIS — HOW IT WORKS

TAN

Proving where a product really comes from — from the factory floor to the buyer's phone

1 THE PEOPLE



Suppliers · web app

Every company in the chain signs off on its own step — what it made, where the work was done, and what it cost — using a secure digital signature.



The chain

Each step is locked to the steps before it, so the whole history of a product is tamper-evident — change one link and it shows.



Buyers · phone app

The **AEGIS Scan** mobile app points the phone camera at the product's QR code and instantly shows where it's really from — no typing, no login.

2 THE APPLICATION



Security gate

Before anything is saved, AEGIS checks the signature is genuine and the supplier is registered. Anything that fails is rejected on the spot.



Supplier & buyer features

Registering companies, recording steps, publishing products, and generating QR codes — all the everyday actions the apps need.



The verifier

The core service that takes a product's history and decides whether it's trustworthy and how Canadian it is.

3 TWO WAYS TO CHECK A PRODUCT

The trust check

Is this history real, and how Canadian is it?

1

Rebuild history

piece the steps together

2

Check signatures

confirm nothing was altered

3

Add up costs

by where work happened

4

Give a label

"Made in Canada," etc.

► **Catches cheating** — forged signatures, broken links, impossible quantities, looping or out-of-order steps, and claims copied from another product.

→ a clear verdict: trustworthy or not, % Canadian, and the label

The buyer's report

A full, readable story of the product

1

Look up history

from the database

2

Handle gaps

still works if data's partial

3

Break down cost

by supplier and step

4

Build report

a "product passport"

■ **Rich detail** — the full supplier chain, a cost breakdown, where the final work was done, and any warnings, all in plain language.

→ a product passport the buyer's app displays

4 THE DATABASE — WHERE EVERYTHING IS STORED



One database, four record types

PostgreSQL



Suppliers

Who's who

Each company and its public signature key — so AEGIS can confirm a step really came from them.



Steps

The claims

Every signed step in a product's history, kept exactly as it was sent so the signature always holds.



Links

What built what

Records which earlier steps fed into each new one — this is what forms the product's full history.



Products

On the shelf

Finished products a supplier has published, ready for a buyer to scan and look up by QR code.